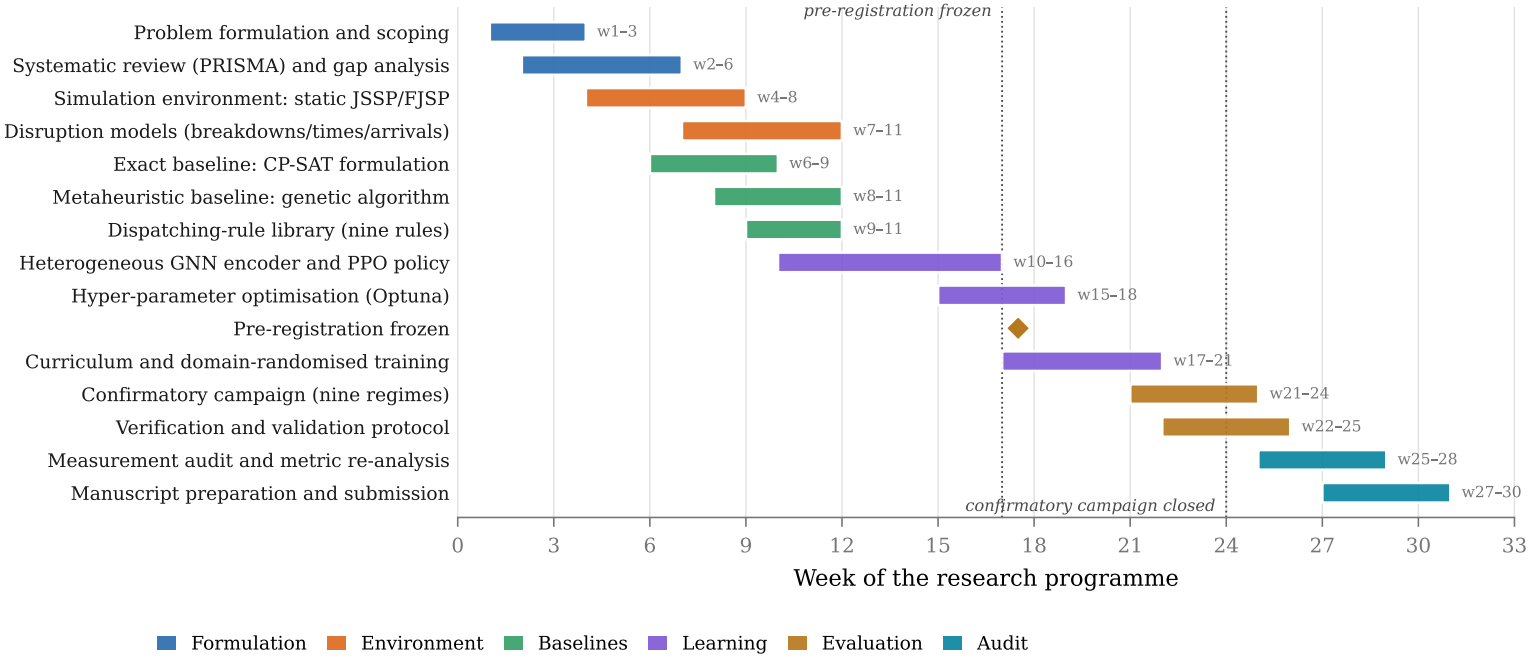
pre-registration frozen
Problem formulation and scoping
w1–3
Systematic review (PRISMA) and gap analysis
w2–6
Simulation environment: static JSSP/FJSP
w4–8
Disruption models (breakdowns/times/arrivals)
w7–11
Exact baseline: CP-SAT formulation
w6–9
Metaheuristic baseline: genetic algorithm
w8–11
Dispatching-rule library (nine rules)
w9–11
Heterogeneous GNN encoder and PPO policy
w10–16
Hyper-parameter optimisation (Optuna)
w15–18
Pre-registration frozen
Curriculum and domain-randomised training
w17–21
Confirmatory campaign (nine regimes)
w21–24
Verification and validation protocol
w22–25
Measurement audit and metric re-analysis
w25–28
Manuscript preparation and submission
w27–30
confirmatory campaign closed
0
3
6
9
12
15
18
21
24
27
30
33
Week of the research programme
Formulation
Environment
Baselines
Learning
Evaluation
Audit